Uses of Electromagnetic Waves

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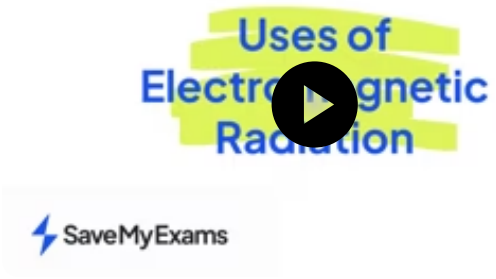

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Uses of electromagnetic waves

Typical uses of the different regions of the electromagnetic spectrum

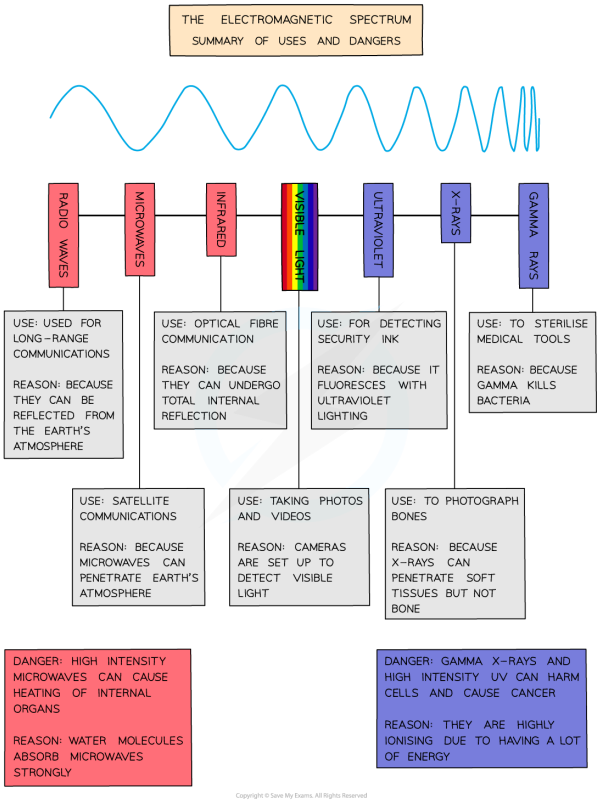
Wave	Use
Radio	Radio and television transmissions Astronomy Radio frequency identification (RFID)
Microwave	Satellite television Mobile (cell) phones Microwave ovens
Infrared	Electric grills Television remote controllers Intruder alarms Thermal imaging

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

Visible light	Vision Photography Illumination
Ultraviolet	Security marking Detecting fake bank notes Sterilising water
X-rays	Medical scanning Security scanners
Gamma rays	Sterilising food Sterilising medical equipment Detection of cancer Treatment of cancer

Typical uses of the different regions of the electromagnetic spectrum



Uses and dangers of the electromagnetic spectrum

Uses of radio and microwaves

- The main uses of radio and microwaves are for wireless communication

- At very high intensities microwaves can also be used to heat things
  - This is what happens in a microwave oven

Uses of infrared

- Warm objects emit infrared radiation
- Infrared can be detected using thermal imaging cameras
  - In security cameras, for research and also in medicine
- Remote controls have small infrared LEDs that send invisible signals to an infrared receiver on a device such as a TV
- Infrared travels down fibre optic cables more efficiently than visible light
  - So most fibre optic communication systems use infrared

Uses of visible light

- Visible light is the only part of the electromagnetic spectrum that the human eye can see
- Visible light is used for photography, videography and illumination

Uses of ultraviolet light

- Ultraviolet is responsible for giving you a sun tan, which is your body's way of protecting itself against the ultraviolet rays
- When certain substances are exposed to ultraviolet, they absorb and re-emit it as visible light (making them glow)
  - This process is known as fluorescence
  - Fluorescence can be used to mark things using special ink secretly
  - Most bank notes have invisible fluorescent markings on them
- Fluorescent light bulbs also use this principle to emit visible light

Uses of X-rays

- The most obvious use of x-rays is in medicine
- X-rays pass through body tissues but are absorbed by denser body parts such as bones
  - Bones absorb X-rays, leaving a shadow which can be seen using photographic film

Uses of gamma rays

- Gamma rays are used to kill cells and living tissue
- This property is utilised in both cancer detection and treatment
  - If gamma rays are carefully aimed at cancerous tissue, they can be effective at destroying the cancerous cells
- Gamma rays can also be used to sterilise food and medical equipment by killing off the bacteria

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